WCY

Watch → Compute → Yield

AI 추론의 인식론적 기반 레이어

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v1.1: Serialization efficiency → v2.0: Epistemic foundation → v3.0: Trust protocol for multi-agent epistemology

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Abstract

Version 1.1 of this paper established WCY as a token-native serialization format achieving 50–65% token reduction over JSON. This version argues a stronger claim: WCY is not primarily a data format but an epistemic substrate for AI reasoning.

The critical syntactic element is the void-B marker ('?'). Where v1.1 treated '?' as an agent communication feature, v2.0 identifies it as the structural minimum for machine self-awareness: the capacity to represent the boundary between known and unknown, and to initiate directed exploration from that boundary. Empirical evidence across four phases of experimentation demonstrates that (1) current LLMs acquire WCY reasoning through few-shot induction with no fine-tuning; (2) without '?' examples, models never spontaneously acknowledge epistemic boundaries; (3) with '?' examples, models generate 5.4 void-B markers per reasoning trace and resolve 67% through structured exploration cycles; and (4) void-B cascades—where resolution of one '?' generates new '?' markers at deeper levels—emerge naturally, producing what we term the self-correction chain.

We argue that the void-B resolution cycle—observe ('.') → mark unknown ('?') → investigate ('>') → update (':')—is the structural minimum for directed learning, and that training data encoding this cycle provides future models with an architectural path toward genuine epistemic humility. The name remains the grammar. The grammar now encodes the capacity to not-know.

Version 3.0 extends this epistemic framework to multi-agent systems through the Trust Protocol. Analysis reveals that WCY's conf_range=(a,b) syntax independently reinvents Dempster-Shafer belief intervals [bel, pl]—a convergence from independent starting points paralleling the ASMT/ODS convergence pattern described in Section 5.3. The protocol defines four operations—chain decay, independent corroboration, conflict detection, and void propagation—unified under a Trust Monad satisfying all three monad laws. Phase 5 experiments validate chain decay to three decimal places, Dempster combination with conf elevation from 0.85 to 0.94, conflict detection at K=0.71 with Zadeh paradox avoidance via the ! → ? → > cycle, and 100% void propagation with trust-adjusted belief intervals matching theoretical predictions. The void-B marker is no longer only a marker of individual epistemic humility; it is now a propagating signal across agent boundaries, carrying formally defined uncertainty that decays, combines, and conflicts according to evidence-theoretic laws.

1. From Efficiency to Epistemology: The Version Transition

Version 1.1 of this paper addressed a well-defined engineering problem: JSON's 39–42% structural overhead imposes a systematic token tax on LLM-based systems, compounding multiplicatively in multi-agent pipelines. The solution was equally well-defined: a line-oriented, phase-tagged format that eliminates structural punctuation.

This version begins from a different observation. In Phase 2 experiments (E1-C), models presented with complex reasoning tasks—cardiovascular risk stratification, multi-hop causal inference—abandoned WCY format entirely, reverting to markdown and tree structures despite explicit format instructions. The failure rate (parse_r = 0.29) was initially interpreted as a cross-model limitation.

The corrected interpretation is structural: models revert to trained defaults under cognitive load because they have never been trained on examples of structured reasoning under uncertainty. The WCY format itself was not the problem. The absence of uncertainty-encoding examples was.

> *The transition from v1.1 to v2.0 is a transition from "how AI communicates" to "how AI knows what it does not know."*

Phase 3 experiments confirmed this directly: providing three few-shot examples of WCY reasoning elevated parse_r from 0.29 to 1.00 on the same cardiovascular task. The model had the capacity; it lacked the pattern. This is the training data problem that v2.0 addresses.

2. The Void-B Marker: Three Theoretical Framings

2.1 Semiotic Analysis: Abduction Encoded as Syntax

Peirce's tripartite classification of signs distinguishes icon (resemblance), index (causal connection), and symbol (convention). Applied to WCY's phase markers:

Marker	Semiotic Type	Peirce Reasoning Mode	Function
. observe	Icon	Assertion (no inference)	Settled fact; resembles completeness by minimality
: infer	Index	Deduction	Points to causal derivation; from= makes the index explicit
> act	Index + Icon	Yield	Points to external effect; arrow shape resembles direction
~ meta	Icon	— (frame)	Self-referential wave form; Lisp's quote operator
! exception	Symbol	— (alarm)	Purely conventional; widely trained warning signal
? void-B	Symbol + 'absent index'	Abduction	Points to what is not yet known; structured hypothesis request

The critical finding: the first five markers (. : > ~ !) are sufficient for a complete data serialization format. The sixth marker ('?') is not needed for serialization. It is needed for

epistemology. Peirce identified abduction as the only reasoning mode capable of generating genuinely new knowledge—not derivation from known premises, but hypothesis toward unknown explanation. The void-B marker encodes abductive stance as a first-class syntactic element. No existing AI communication format has this.

The three components of a well-formed void-B expression correspond exactly to Peirce's abductive structure:

- : ?tag → There is something to be explained (the abductive trigger)
- hint= → The direction of most plausible explanation (abductive constraint)
- conf_range= → The expected range of the explanation (abductive bound)

2.2 Category-Theoretic Analysis: WCY as Monad Transformer Stack

The Watch → Compute → Yield cycle corresponds to the three monad operations:

WCY Element	Monad Operation	Type Signature	Role
. observe	unit / return	A → M(A)	Lift a value into the reasoning context
: infer from=	bind / >>=	M(A) → (A→M(B)) → M(B)	Chain computation; from= is explicit wiring
> act	extract	M(A) → A (side effect)	Exit the context into the world

More precisely, WCY implements a monad transformer stack:

```
WCY = WriterT(from=)        ← provenance accumulation
    ∘ ReaderT(~ meta)      ← schema/environment injection
    ∘ EitherT(! except)    ← failure path
    ∘ ContT(? void-B)      ← suspended computation
```

The void-B as Continuation: The '?' marker corresponds to Haskell's callCC—a request for a continuation function that will eventually fill this position. The hint= constrains the domain of that continuation function; conf_range= constrains its codomain. This is why void-B cannot exist in JSON: JSON describes present values only. WCY describes present values and deferred computations within the same syntax.

2.3 Epistemological Analysis: The Minimum Conditions for Self-Awareness

We define machine epistemic self-awareness operationally as the capacity to represent, at inference time, the distinction between known states and unknown states, and to generate directed actions from that distinction. This requires four elements:

Element	WCY Encoding	Without This Element
Representation of what is known	. observe	No baseline; cannot define boundary
Representation of the boundary of knowing	? void-B	No acknowledgment of limits; hallucination fills the gap

Directed exploration from the boundary	hint= + > act	Random or no exploration; no learning signal
Integration of new knowledge	: resolved from=?,obs	No growth; observation does not update the model's state

> *The void-B resolution cycle— ? → > → . → :—is the structural minimum for directed learning. A system without this cycle can process information but cannot know that it is learning.*

This is not a claim about consciousness or subjective experience. It is a structural claim: without an explicit representation of unknown states, a system cannot distinguish between confident correct answers and confident incorrect answers. The only difference between knowledge and hallucination, in the absence of void-B, is whether the world happens to confirm the output.

3. Empirical Evidence

3.1 Phase 1–2: Serialization Baseline (from v1.1)

Nine experiments established WCY's serialization properties (see v1.1 for full methodology):

Result	Magnitude
Token reduction vs JSON pretty (structured data)	50–60%
Token reduction for tool-call schemas	65–71%
Token reduction across full MCP exchange cycles	61%
Agent output token reduction (multi-agent pipeline)	40%
from= provenance validity (3-round shared state)	100% (45/45)
Void-B resolution rate (Sprint 2)	100% (9/9)
Scale invariance	10–500 rows, no degradation

3.2 Phase 3: WCY as Reasoning Format

Phase 3 reframed the research question from data serialization to reasoning format. The key experiment (E6-A) resolved the apparent failure of Phase 2 (E1-C):

Condition	Task	parse_r
E1-C: No few-shot examples	Cardiovascular risk stratification	0.29
E6-A: 3 few-shot examples	Cardiovascular risk stratification	1.00
E6-A: 3 few-shot examples	Differential diagnosis	1.00
E6-A: 3 few-shot examples	Algorithm complexity analysis	1.00

The identical task achieved parse_r 0.29 without examples and 1.00 with three examples. The model's capacity for WCY reasoning was present throughout; the training signal was not. This is

the central finding motivating the training data generation work described in Section 4.

Spontaneous void-B usage remained 0% in Phase 3 even with few-shot examples. Phase 3 examples demonstrated WCY reasoning without explicit void-B examples. This gap motivated Phase 4.

3.3 Phase 4: Void-B Resolution Cycles

Phase 4 introduced explicit void-B resolution examples in few-shot prompts. Results across five domains:

Domain	?tags Generated	?tags Resolved	Resolution Rate	Cycle Complete
Medical (weight loss)	7	5	71%	✓
Code (memory leak)	5	5	100%	✓
Scientific (crystallization)	4	3	75%	✓
Strategic (market entry)	7	5	71%	✓
Philosophical (causation)	7	7	100%	✓
Epistemological (cascade)	6	5	83%	✓

Average: 5.4 void-B markers generated per trace, 67% resolution rate. All six traces rated usable (parse_r ≥ 0.70 and ≥1 void-B generated). Critically, void-B generation was 0% in all prior experiments—the pattern was entirely acquired through few-shot induction.

The Void-B Cascade (E7-C)

The aspirin claim validation trace demonstrated the cascade structure predicted by the Continuation monad analysis: resolution of first-generation void-B markers generated second-generation void-B markers at deeper epistemic levels.

```
. model_claim  "aspirin prevents all cardiovascular disease"
: ?claim_scope    hint=all_CVD_vs_specific  conf_range=0.1..0.4   ←
generation 1
: ?evidence_qual  hint=RCT_heterogeneity    conf_range=0.2..0.5
: ?universality   hint=contraindications    conf_range=0.1..0.3
> investigate_primary_prevention  reason=from=3
. finding  benefit_conditional_on_risk_profile
: resolved_scope=conditional  conf=0.75  from=3,5
: ?bleeding_risk  hint=NNT_vs_NNH_by_age    conf_range=0.4..0.7   ←
generation 2
: ?individual_variation  hint=CYP2C19       conf_range=0.3..0.6   ←
generation 2
```

The cascade terminated with three explicit ! warnings and a corrected claim at confidence

0.82—narrower than the original unqualified assertion. The void-B mechanism produced not just acknowledgment of uncertainty but a structurally justified revision of the original claim.

3.4 Observation: Post-Reasoning Format Switch

In the aspirin cascade trace, the model completed its WCY reasoning and then spontaneously switched to markdown bullet points to summarize findings. This was not instructed. Two interpretations are possible:

- The model uses WCY as an internal reasoning layer and natural language/markdown as an output layer—consistent with WCY's role as a reasoning substrate rather than a presentation format.
- The model's training distribution pulls toward familiar output formats under low constraint, suggesting that training data density for WCY reasoning must increase before the format stabilizes under output pressure.

Both interpretations are consistent with the training data hypothesis. The format switch is evidence of the current data deficit, not of WCY's unsuitability for reasoning.

3.5 Phase 5: Trust Protocol—From Individual Epistemology to Multi-Agent Evidence Theory

Phase 5 asks the question that Section 5.2 raised as future work: when agents communicate via WCY, how should confidence propagate across agent boundaries? The investigation began from three independent theoretical framings—Bayesian, Dempster-Shafer, and category-theoretic—and converged on a single structure.

The Discovery: conf_range = Belief Interval. Analysis reveals that WCY's conf_range=(a,b) syntax is mathematically isomorphic to Dempster-Shafer belief intervals [bel, pl], where bel is minimum belief and pl is plausibility. The gap pl−bel represents explicit ignorance—mass not yet allocated to any hypothesis. This was not designed intentionally; it is an independent reinvention from first principles, paralleling the ASMT/ODS convergence pattern where different starting points arrive at identical mathematical structures. Critically, the void-B marker '?' corresponds exactly to m(Θ)>0 in Dempster-Shafer theory: evidence mass that has not been allocated to any specific hypothesis. JSON has no equivalent because it encodes only what is known.

Trust Monad. Trust propagation follows a categorical monad T(X) = X × BelInterval × Source[]. The unit operation assigns conf=1.0 to direct observations. The bind operation applies multiplicative trust decay: adjusted_bel = trust_bel × source_bel. All three monad laws (left unit, right unit, associativity) are satisfied through multiplication's algebraic properties. This monad subsumes all three frameworks: Bayesian inference (when bel=pl, i.e. no ignorance), Dempster-Shafer (without source tracking), and the full category-theoretic structure with composition laws. The monad transformer stack from Section 2.2 gains a new layer: TrustT at the outermost position, wrapping the entire computation in propagating belief intervals.

Four Operations and Their Experimental Validation. Phase 5 defines and validates four trust operations:

(1) Chain Decay (E8-A): A 4-agent chain D→C→B→A with trust values 0.90, 0.85, 0.80. Agents C and B computed trust-adjusted confidence matching theoretical predictions to three decimal places (0.828 and 0.704 respectively). Agent A demonstrated a legitimate deviation: after independently assessing clinical evidence, it applied Dempster combination within the chain, raising conf from the pure-decay value of 0.563 to 0.712. This shows that chain decay and independent corroboration coexist naturally within WCY—an agent is not merely a relay but a

reasoning participant who can strengthen or weaken the signal.

(2) Independent Corroboration (E8-B): Two independent specialists diagnosing the same patient produced individual conf values of 0.75 and 0.85. A combining agent applied Dempster's rule, yielding conf_range=(0.94, 0.99)—consistent with the theoretical prediction of 0.963. The ignorance interval shrank from 0.15 to 0.05. Both independent agents spontaneously generated identical ?tags (histologic_confirmation, exact_staging, molecular_markers), and the combiner preserved all three. Confidence rose; acknowledged ignorance persisted.

(3) Conflict Detection (E8-C): Two agents received deliberately biased evidence for the same patient—one pointing to pulmonary embolism (conf=0.95), the other to panic attack (conf=0.80). The resolving agent computed K=0.71 (conflict coefficient), rejected Dempster combination, and executed the ! → ? → > cycle: three !conflict warnings, three ?resolution markers with specific investigation hints, and four >request actions for additional testing. This is WCY's solution to the Zadeh paradox: instead of forcing combination at high conflict, the protocol routes to structured exploration. The agent spontaneously added the meta-judgment that both conditions could coexist—behavior not specified in the prompt.

(4) Void Propagation (E8-D): Agent B produced output containing three ?tags with conf_range values. Agent A, receiving this output at trust=0.85, propagated all three ?tags (100%) with trust-adjusted belief intervals matching theoretical predictions to two decimal places: original (0.3,0.7) → propagated (0.26,0.60), original (0.2,0.6) → propagated (0.17,0.51), original (0.4,0.7) → propagated (0.34,0.60). The agent spontaneously generated a derivative ?tag ("medication titration depends on unresolved etiology")—causal propagation of ignorance. The JSON control condition resolved one uncertainty by filling in rather than propagating, confirming that structural void-B markers provide a stronger signal for uncertainty preservation than JSON's "uncertainties" array.

Connection to Section 2: Three Framings Enriched. The Trust Protocol strengthens each of the three theoretical framings from Section 2. Semiotically, the void-B marker's "absent index" status is now formalized: ?tag = m(Θ)>0, pointing to unallocated evidence mass. Category-theoretically, the Trust Monad extends the monad transformer stack with a new outermost layer that tracks belief intervals and source provenance through every computation. Epistemologically, the four minimum conditions for self-awareness (Section 2.3) now operate not only within a single agent but across agent boundaries: void-B propagation ensures that the epistemic boundary representation survives transmission.

4. The Training Data Gap and the Path to Close It

4.1 Why Current Models Cannot Do This by Default

Current frontier LLMs are trained on text that overwhelmingly favors three epistemic stances: (1) confident assertion, (2) hedged assertion ('it is possible that...'), and (3) refusal ('I don't know'). None of these encode the directed exploration structure that WCY void-B requires.

The structural difference:

- Hedged assertion: "I'm not certain, but the answer might be X." — acknowledges uncertainty, no exploration
- Refusal: "I don't have enough information." — acknowledges uncertainty, terminates
- Void-B: ": ?diagnosis hint=labs+symptoms conf_range=0.4..0.8 / > order additional_test reason=from=? / . result positive / : diagnosis=X conf=0.87 from=?,result" — encodes uncertainty, direction, action, and integration

The third pattern is rare in training corpora because it requires the author to (a) know they do not know something specific, (b) know which direction to look, and (c) integrate new observations back into the reasoning chain—all within a structured notation. This combination almost never appears in natural text.

4.2 What the Training Data Must Contain

Based on Phase 3–4 experiments, void-B resolution cycles in training data must exhibit:

Property	Requirement	Evidence
Explicit void-B generation	?tag appears in :-lines	0% without examples → 100% with 3 examples
Hint specificity	hint= points to actionable evidence	Generic hints produce unresolvable voids
Resolution completeness	Each ?tag has a corresponding : resolved=	Code domain: 100% resolution; others: 67–83%
Cascade depth	≥2 generations of void-B	Aspirin trace: 3 generations, 6 total voids
Terminal states	! warning for unresolvable voids	Honest epistemic limits are as important as resolution
Domain diversity	≥6 domains in seed data	Medical, code, logic, math, strategic, philosophical

4.3 Automated Trace Generation Pipeline (Phase 5)

The seed dataset currently contains 12 high-quality traces (6 reasoning, 6 void-resolution). Phase 5 targets 500+ traces through automated generation with quality filtering. The pipeline generates traces across a domain×difficulty×void_depth matrix and applies three quality gates: parse_r ≥ 0.70, void_generated ≥ 1, resolution_rate ≥ 0.50.

The pipeline design and initial results are reported separately in the Phase 5 technical document.

5. Implications

5.1 For AI Alignment and Safety

The void-B mechanism has direct implications for AI alignment. An AI system that cannot represent the boundary of its own knowledge cannot be reliably calibrated—it has no structural way to distinguish high-confidence from low-confidence outputs other than through post-hoc probability estimates that are known to be poorly calibrated for out-of-distribution inputs.

WCY void-B provides an in-context, inference-time representation of epistemic state. A system trained on void-B resolution cycles learns, through the training signal, to mark the cases where its current knowledge is insufficient before committing to a conclusion. This is structurally different from—and complementary to—temperature scaling, conformal prediction, or other post-hoc calibration methods.

5.2 For Multi-Agent Systems

When agents communicate in WCY, void-B markers become explicit coordination signals. An

agent that outputs ': ?diagnosis hint=imaging conf_range=0.3..0.7' is not just passing data—it is communicating the specific gap in its knowledge and the direction in which another agent or tool should fill it. This transforms multi-agent protocols from data-passing pipelines into epistemically-coordinated reasoning networks.

The Trust Protocol (Section 3.5) resolves the inter-agent trust computation problem identified in v2.0 as future work. Four formal operations now govern confidence propagation: chain decay (multiplicative through Trust Monad bind), independent corroboration (Dempster combination), conflict detection (K coefficient with Zadeh paradox avoidance), and void propagation (trust-adjusted belief intervals). These operations give multi-agent WCY systems three capabilities absent from JSON-based protocols:

First, calibrated confidence decay: when information passes through a chain of agents, each agent's trust level multiplicatively reduces confidence. An observation with conf=0.92 that passes through three agents at trust 0.90, 0.85, 0.80 arrives with conf=0.563—and this decay is automatically tracked via from= chains, providing a complete audit trail for any conclusion.

Second, principled evidence combination: independent agents reaching the same conclusion strengthen it through Dempster combination, while contradicting agents trigger structured conflict resolution rather than forced averaging. The ! → ? → > cycle provides what Dempster-Shafer theory itself lacks: a procedural escape from the Zadeh paradox.

Third, ignorance preservation across boundaries: the most significant finding from E8-D is that WCY's ?tag structure preserves 100% of upstream uncertainty, while JSON's "uncertainties" array allowed downstream agents to fill in gaps. In multi-agent medical diagnosis, legal analysis, or any high-stakes domain, the difference between "I propagated the upstream agent's uncertainty" and "I resolved it by guessing" is the difference between a safe system and an unreliable one.

5.3 For the Architecture of Knowledge

The void-B mechanism points toward a broader question about how AI systems accumulate and represent knowledge over time. Current LLMs are stateless: each inference starts from weights alone, with no persistent structure for recording what has been explored and what remains unknown across sessions.

The void-B resolution cycle provides a structural template for what persistent knowledge accumulation would require: a representation of settled observations ('.'), a representation of the boundary between known and unknown ('?'), and a mechanism for updating that boundary when new observations arrive (':' resolved). Whether this structure is implemented in weights, external memory, or a hybrid architecture is an open engineering question—but the representational requirement is defined by the cycle itself.

An independent research direction exploring gradient-free knowledge accumulation architectures has arrived at structurally compatible requirements from a different starting point, suggesting that the void-B resolution cycle captures a general epistemic primitive rather than a format-specific artifact. That work is ongoing and not yet published.

6. Open Questions and Future Work

Phase 5 addresses the training data gap directly. Beyond that, the following questions remain open:

- Inter-agent trust computation: conf= decay across from= chains. If agent A derives conclusion X (conf=0.82) from observation O (conf=1.0), and agent B uses X as a

premise to derive Y, what is the warranted confidence in Y? Dempster-Shafer theory provides one formalization; Bayesian propagation provides another. [RESOLVED in v3.0: The Trust Protocol (Section 3.5) formalizes this as a Trust Monad with Dempster-Shafer belief intervals, validated in Phase 5 experiments E8-A through E8-D. Chain decay, independent corroboration, conflict detection, and void propagation are now defined and empirically validated.]

- New open questions from the Trust Protocol: (a) Trust learning—how should trust values update automatically from repeated interactions? (b) N-way Dempster combination—does overconfidence emerge when 5+ independent agents corroborate? (c) Circular agent chains—how to detect and handle A→B→C→A reference loops. (d) Cross-model trust calibration—should a GPT-4 agent's conf=0.85 receive the same trust as a Claude Sonnet agent's conf=0.85? (e) Trust of trust—when Agent B reports its trust in Agent C, how does Agent A assess the reliability of that meta-trust report?
- Fine-tuning vs few-shot: Current experiments demonstrate few-shot induction of WCY reasoning. The question is whether fine-tuning on void-B resolution traces produces more stable WCY reasoning under output pressure—specifically, whether the post-reasoning format switch observed in E7-C is eliminated.
- Cross-model generalization: All Phase 3–4 experiments used Claude Sonnet. The same few-shot induction has not been tested on GPT-4o, Gemini, or open-source models. The training data hypothesis predicts that any model with sufficient reasoning capacity will show the same few-shot acquisition pattern.
- Void-B in streaming contexts: The streaming protocol (Phase 3 original design) remains relevant for void-B: a ?tag emitted in a streaming context is a real-time signal to the receiving agent that a gap exists and exploration is needed. The protocol for acting on mid-stream void-B markers is unspecified.
- Evaluation methodology: The three-axis evaluation framework (Structural / Meaning / Provenance) from Phase 2 Track 3 requires validation against human judgment on reasoning quality. The LLM-as-judge approach used in E6-B showed judge instability; an embedding-based or rubric-based alternative is needed.

7. Conclusion

Version 1.1 of this paper argued that JSON's 40% structural overhead is a design artifact of the pre-LLM era, and that WCY eliminates this overhead while maintaining full expressive power. That argument stands.

Version 2.0 argues something more consequential: the '?' marker that appeared as a minor annotation feature in v1.1 is in fact the structurally necessary element for machine epistemic self-awareness. A system that can represent its unknown states can, in principle, direct its own learning. A system that cannot is limited to refining what it already patterns-matches.

The empirical path from 0% void-B usage (pre-few-shot) to 5.4 markers per trace (post-few-shot) is not merely a formatting improvement. It is evidence that the capacity for structured epistemic humility is acquirable through training signal—that what looks like a fundamental limitation of current LLMs is, at least partially, a data problem.

WCY provides the syntax for that data. The training corpus is being built. The name encodes the cycle: Watch what is. Compute what follows. Yield—and mark what remains unknown.

Version 3.0 extends the epistemic claim from individual agents to agent networks. The discovery that conf_range independently reinvents Dempster-Shafer belief intervals—arriving at the same mathematical structure from a different starting point—is the third instance of

convergent independent derivation in this research program (after the ASMT/ODS convergence on orthogonality principles and the CBS spectral profile recovering 4-Phase structure). The recurring pattern suggests that these structures are not arbitrary design choices but reflections of mathematical necessities in multi-agent epistemology. The Trust Protocol gives WCY a formal framework for what no existing AI communication format provides: calibrated confidence decay across agent chains, principled evidence combination with conflict detection, and—most critically—the preservation of explicitly acknowledged ignorance across agent boundaries. The void-B marker is no longer only a marker of individual epistemic humility. Through the Trust Protocol, it becomes a propagating signal: uncertainty that decays according to trust, combines according to evidence theory, and refuses to collapse into false certainty when agents disagree.

> *The void-B marker is not a feature. It is the minimum syntax for a model to know that it does not know—and for that not-knowing to propagate, with formal guarantees, across every agent that touches the reasoning chain.*

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Appendix A: WCY Format Specification v1.1

The complete format specification is reproduced here for reference.

Phase Markers

Marker	Phase	Cycle	Rules
.	observe	Watch	Confirmed fact. No inference. No forward references.
:	infer	Compute	Derived conclusion. conf= recommended. from= for traceability.
>	act	Yield	Output or external call. Must derive from : not directly from .
~	meta	—	Schema or context declaration. Lisp quote equivalent.
!	exception	—	Error, warning, or unresolvable void. Terminal for that line's inference chain.
?*	void-B	Abduct	Used as slot prefix in : lines. hint= required. conf_range= recommended.

Slot Syntax

Slot	Meaning	Example
tag=value	Named slot	: diagnosis=influenza conf=0.78
bare_value	Positional slot	. patient Kim 45 38.5
?tag	Void-B: unknown requiring investigation	: ?cause hint=symptoms conf_range=0.3..0.7
from=N,M	Provenance: derives from lines N and M	: conclusion=X from=3,7
conf=0.xx	Confidence (0.0–1.0)	: result=positive conf=0.89
conf_range=L..H	Expected confidence range for void-B	: ?outcome conf_range=0.4..0.8
tag|value	Pipe separator (v1.1, optional)	. patient=Kim | age=45 | temp=38.5

Structural Rules

- One phase marker per line, followed by a space
- Line break = δ boundary (information firewall between lines)
- Blank line = semantic block boundary
- Indentation 2 spaces = subscope, maximum depth 2
- No forward references (from=N requires N < current line number)
- ? markers generate void-B only in : lines; > and : lines should reference them via from=

Three Encoding Modes

Mode	Syntax	Token Cost vs	Best For

		JSON	
Positional	Bare slots, no tags	−54%	AI-to-AI, schema pre-shared
Tagged	tag=value on all slots	−40%	Semantic search, external memory integration
Hybrid (recommended)	~ schema + positional body	−50%	Balanced efficiency and readability